6. 外围设计原理图

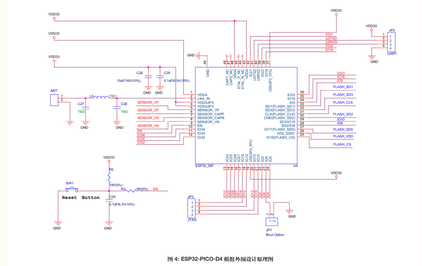
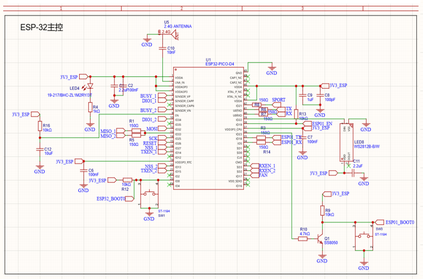
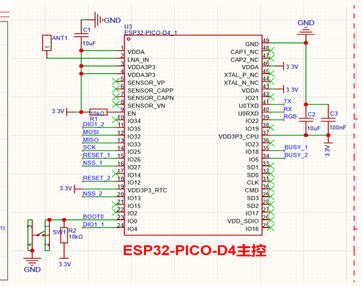
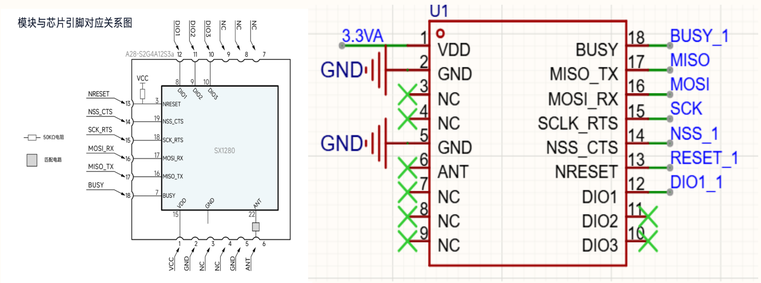
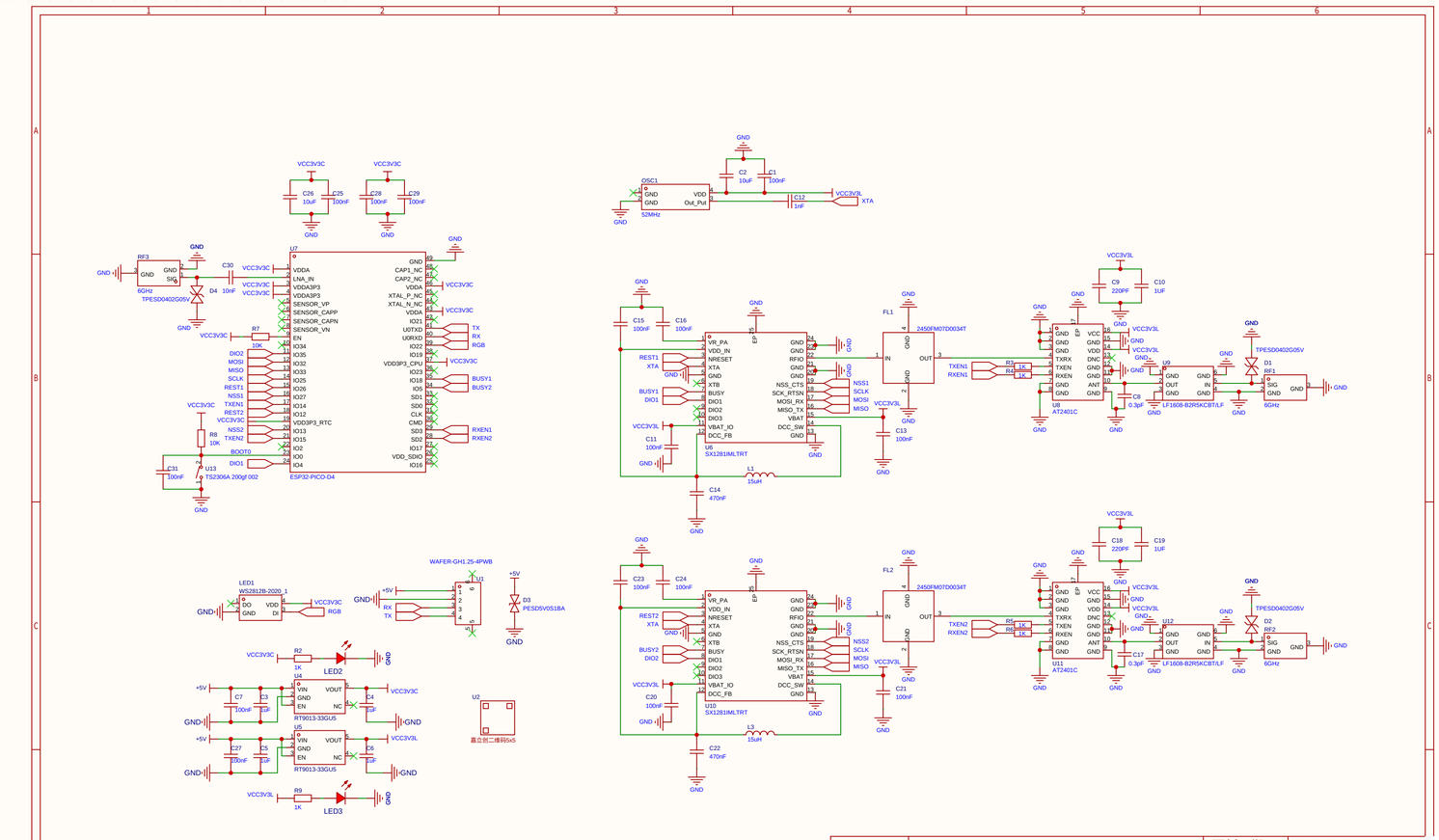


图 4: ESP32-PICO-D4 模拟外围设计原理图

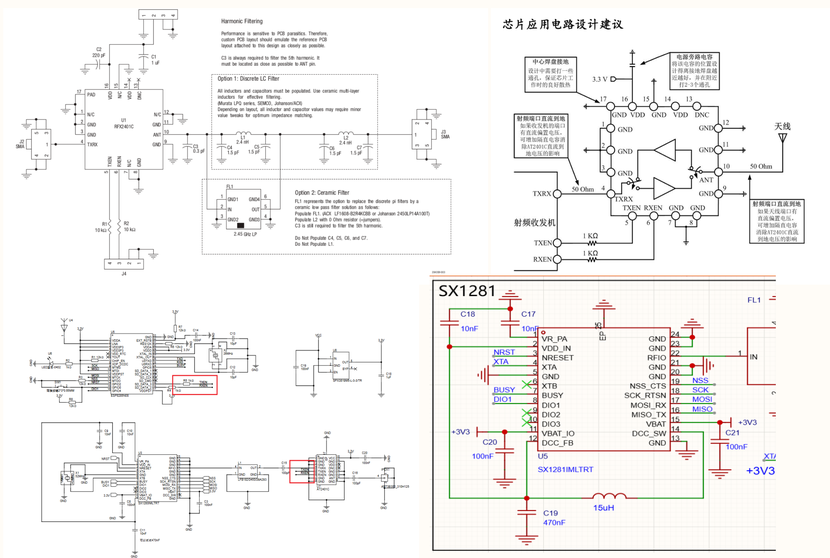


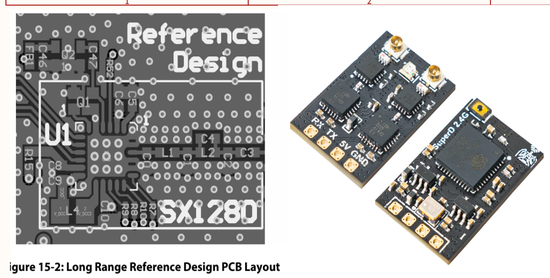






原理图	Gemini	更新日期	2026-01-31
		创建日期	2026-01-31
图页	Gemini	物料编码	
绘制		NxtPX4-CM4	
审阅			
	版本	尺寸	页 1 共 1
嘉立创EDA	V1.0	A3	嘉立创EDA





igure 15-2: Long Range Reference Design PCB Layout

15.2 Application Design with optional TCXO

Although the modulation parameters of the transceiver are designed to tolerate typical frequency drifts associated with the use of industry standard tolerance crystal reference oscillator components, an external Temperature Compensated Crystal Oscillator (TCXO) can be used. The figure below shows how the TCXO should be AC-coupled to the XTA input of transceiver. The TCXO should be of the 0.8 V clipped sine output type. Please consult the crystal manufacturer for full details of the components required specific to your TCXO.

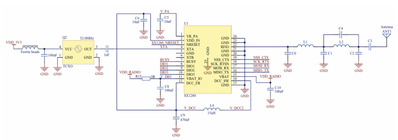
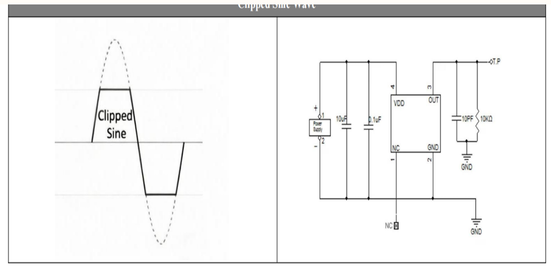


Figure 15-3: Application Schematic with Optional TCXO



三、将削波正弦波形与振荡器IC结合使用

具有削波正弦波输出的 TCXO 也可以与振荡器 IC 结合使用，为芯片组生成方波输出。

- TCXO的输出需要连接到振荡器IC上的晶体输入引脚，振荡器IC的输出引脚应连接到芯片组上的输入引脚。
- 除 7x5mm 产品线外，对于所有其他较小尺寸的 TCXO，需要在 TCXO 输出引脚和 IC 时钟输入引脚之间串联一个 1000 pF 电容器。

1. 参数含义

`10kΩ /10pF` **不是让您外接这两个元件**，而是**规格书中对TCXO输出端负载条件的规定**：

- **10kΩ**：表示TCXO输出端应该看到的**直流负载阻抗**（最小10kΩ）
- **10pF**：表示TCXO输出端允许的**最大容性负载**（包括PCB走线电容+芯片输入电容）

这意味着：您的电路设计必须满足：

1. **直流负载 ≥ 10kΩ**（通常芯片输入阻抗远大于此，自动满足）
2. **总容性负载 ≤ 10pF**（需要计算和控制）

Pin Description-Clipped sine wave（脚位说明-Clipped

Pin	#1	#2	#3	#4
Function	No Connect	Ground	Output	Supply Voltage

Notes:
1.Please keep the NC pin floating during use. Do not connect it to any network (power, ground, or signal).
2.The provided pinout is for reference purposes only. The definitive material specification document must be obtained by contacting a sales representative.